CRISTOBAL, HAILEY JADE BSIT 2-1 Activity #1 - Variables OOP 3/2/26

PART A → VARIABLE IDENTIFICATION TABLE

VARIABLE NAME	DATA TYPE	VARIABLE TYPE	STORED IN
s1	Student	Reference Variable	Stack
s2	Student	Reference Variable	Stack
s3	Student	Reference Variable	Stack
name	String	Instance Variable	Heap
grade	int	Instance Variable	Heap

PART B → Explanation

STATEMENT	WHAT HAPPENS?	MEMORY IMPACT
Student s1 = new Student ("John", 85);	We created a new Student object	A new object is allocated on the Heap. s1 on the stack points to it.
Student s2 = new Student ("Maria", 90);	We create a second Student object	Another object is allocated on the heap. s2 points to it.
Student s3 = s2;	s3 is assigned the reference held by s2.	s3 now points to the same object as s2 on the Heap. No new object is created
s1.grade = 95;	The grade of the first student is updated	The value of grade in the object pointed to by s1 changes from 85 to 95
s3.name = "Ana";	The name of the object s3 points to is updated	Since s3 and s2 point to the same object, the name for both changes to "Ana".

PART C → OUTPUT

The output will be:

John 95 Ana 90 Ana 90